Lecture 5 v2:
Multivariate Linear Regression

25117 - Econometrics

Universitat Pompeu Fabra

October 16th, 2024

What we learned in the last lesson

- Hypothesis testing for regression coefficients is analogous to hypothesis testing for the population mean: Use the t-statistic to calculate the p-values and either accept or reject the null hypothesis. Like a confidence interval for the population mean, a 95% confidence interval for a regression coefficient is computed as the estimator ± 1.96 standard errors.

- When X is binary, the regression model can be used to estimate and test hypotheses about the difference between the population means of the "X = 0" group and the "X = 1" group.

- In general, the error u_i is heteroskedastic; that is, the variance of u_i at a given value of X_i, $var(u_i \mid X_i = x)$, depends on x. A special case is when the error is homoskedastic; that is, when $var(u_i \mid X_i = x)$ is constant.

  Homoskedasticity-only standard errors do not produce valid statistical inferences when the errors are heteroskedastic, but heteroskedasticity-robust standard errors do.

What we learned in the last lesson

- If the three least squares assumption hold and if the regression errors are homoskedastic, then, as a result of the Gauss–Markov theorem, the OLS estimator is BLUE.

- If the three least squares assumptions hold, if the regression errors are homoskedastic, and if the regression errors are normally distributed, then the OLS t-statistic computed using homoskedasticity-only standard errors has a Student t distribution when the null hypothesis is true. The difference between the Student t distribution and the normal distribution is negligible if the sample size is moderate or large.

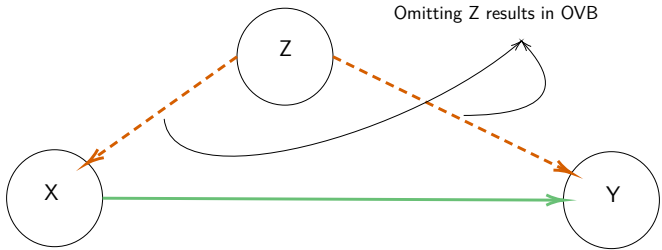
Omitted Variable Bias (OVB)

- If β_1 is a causal effect, the first assumption of the Gauss-Markov Theorem must hold : $\mathbb{E}(u_i \mid X_i) = 0$ for all i
- Remember that $\hat{\beta}_1 = \beta_1 + \frac{cov(X,u)}{var(X)}$ and $\hat{\beta}_1$ will be unbiased only if $\mathbb{E}(\frac{cov(X,u)}{var(X)}) = 0$
- **Example:**
  - Private university graduates tend to have higher wages than public university graduates. Does that mean that private university education (X_1) yields higher wages (Y)?
  - Maybe, but not necessarily — Private university graduates are *selected* by those institutions
  - They differ in a number of ways from public university graduates: they have on average better test scores in highschool, they come from wealthier families, etc.
  - All these relevant but unobserved variables may be omitted, and, if not accounted for, may bias the estimation of β_1
  - In comparing wages from private and public university graduates, how to *identify* the causal effect of education (X_1) and differentiate it from the effect of, say, advantageous social background?
  - We want to make apple-to-apple comparisons...

Omitted Variable Bias (OVB)

The bias in the OLS estimator that occurs as a result of an omitted factor, or variable, is called omitted variable bias. For omitted variable bias to occur, the omitted variable Z must satisfy two conditions:

1. Z is a determinant of Y (i.e. Z is part of u); and
2. Z is correlated with the regressor X (i.e., $corr(Z, X) \neq 0$)

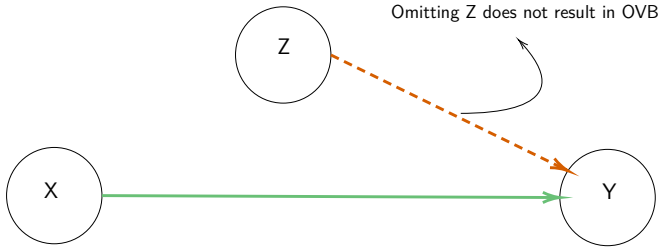


Both conditions must hold for the omission of Z to result in omitted variable bias.

Omitted Variable Bias (OVB)

The bias in the OLS estimator that occurs as a result of an omitted factor, or variable, is called omitted variable bias. For omitted variable bias to occur, the omitted variable Z must satisfy two conditions:

1. Z is a determinant of Y (i.e. Z is part of u); and
2. Z is correlated with the regressor X (i.e., $corr(Z, X) \neq 0$)

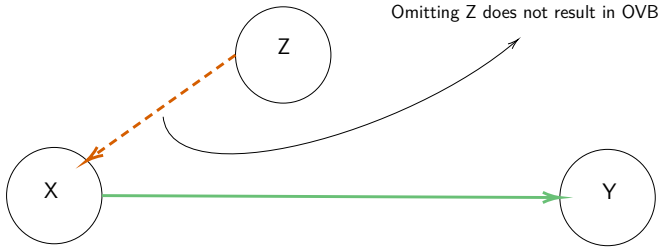


Both conditions must hold for the omission of Z to result in omitted variable bias.

Omitted Variable Bias (OVB)

The bias in the OLS estimator that occurs as a result of an omitted factor, or variable, is called omitted variable bias. For omitted variable bias to occur, the omitted variable Z must satisfy two conditions:

1. Z is a determinant of Y (i.e. Z is part of u); and
2. Z is correlated with the regressor X (i.e., $corr(Z, X) \neq 0$)

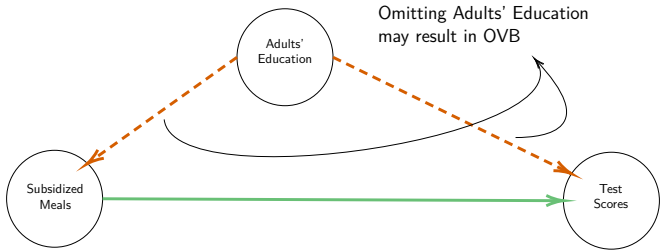


Both conditions must hold for the omission of Z to result in omitted variable bias.

Omitted Variable Bias (OVB)

In the California Schools example,

1. Adults' educational attainment may affect local kids' test scores; and
2. Adults' educational attainment is a driving force of local income, and, therefore, likely affects the share of subsidized meals



Again, both conditions must hold for the omission of Z to result in omitted variable bias.

Omitted Variable Bias (OVB)

There are systematic differences in educational attainment across districts with low and high share of subsidized meals.

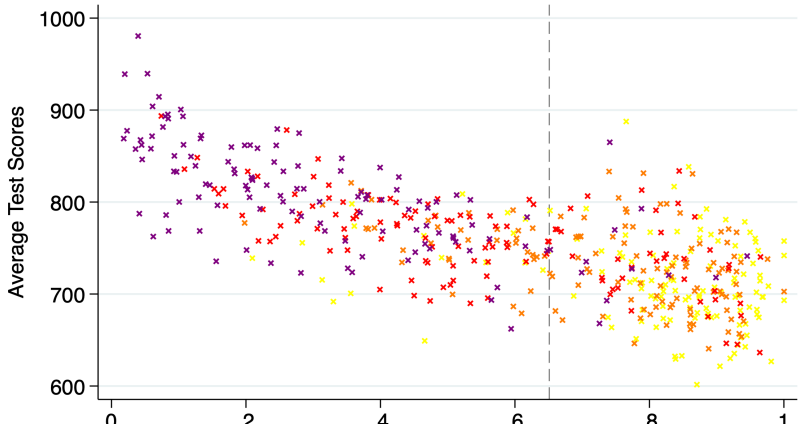
- Districts more educated adults have higher test scores
- Districts less educated adults have more schools with subsidized meals
- Among districts with comparable adults' educational attainment, the effect of subsidized meals is generally smaller

	Above-median % Subsidized meals		Below-median % Subsidized meals		Difference in Test Scores
	$\overline{Y}$	n	$\overline{Y}$	n	$\overline{Y}^1 - \overline{Y}^0$
All Districts	720.2472	250	787.4612	250	67.214
By Share w/ College Education:					
[0; .1339436]	710.3194	108	745.3444	18	35.025
[.1350751; .1996677]	724.0533	90	758.4917	36	34.43834
[.1997542; .3260502]	731.9475	40	774.4559	84	42.50845
[.3309678 ; .7696404]	742.05	12	813.2955	112	71.24553

Another way to see this

The scatterplot shows one point per school:

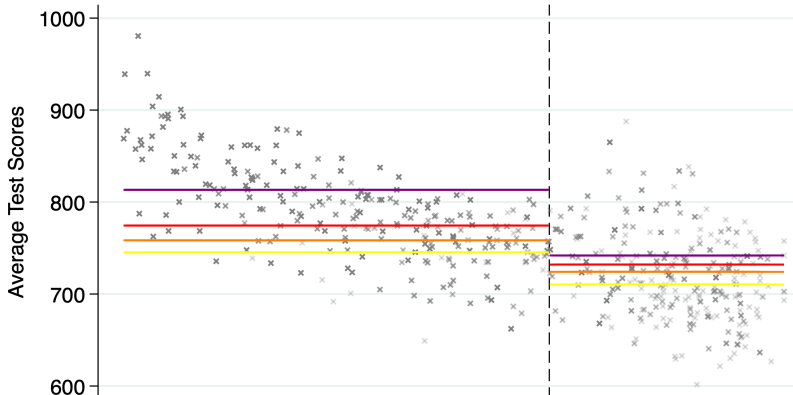
- Horizontal axis: share of subsidized meals (X); vertical axis: average test score (Y).
- Each color represents a quartile of the share of adults with a college education.
- Comparing all points mixes schools with different education levels, so the raw relationship can combine the meal effect with differences in background characteristics.



Another way to see this

The colored horizontal lines are group-specific average test scores:

- Within each education quartile, the line summarizes the average score for schools with a given level of subsidized meals.
- The vertical dashed line separates the above-median and below-median subsidy groups.
- The lines show that education level shifts test scores, so comparing the two subsidy groups without controlling for education is not an apples-to-apples comparison.



What to do about it?

- Randomization implies that any differences between the treatment and control groups are random — not systematically related to the treatment
- From now on, one of the main goals of this course is to explore ways to recover causal treatment effects
- Remember, we want to make ceteris paribus, apple-to-apple comparisons — not pear-to-apple, etc.
- One way to do it is to **control** for systematic differences between control and treatment group
- In our example, we want to know the average treatment effect of subsidizing school meals on test scores, holding the share of college education, average income, English proficiency, etc. constant (i.e., getting closer to all else equal, or ceteris paribus)
- Said differently, we want to know the average impact of a change in subsidized meals on test scores within groups where college education, average income, English proficiency, etc. is the same

The Multiple Regression Model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + ... + \beta_k X_{ki} + u_i \qquad \text{with} \quad i = 1, \ldots, n$$

Where

- Y_i is the ith observation of the dependant variable
- $X_{1i}, X_{2i}, ..., X_{ki}$ is the ith observation of the kth regressor
- u_i is the error term

The population regression line is therefore given by:

$$\mathbb{E}(Y_i \mid X_{1i} = x_1, X_{2i} = x_2, ..., X_{ki} = x_k) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k$$

- β_k is the slope of X_{ki}
- β_k is the expected difference in Y_i associated with a unit change in X_{ki} holding all other regressors constant
- β_0, the intercept, can be thought of as the coefficient on a regressor X_{0i} that is equal to 1 for all observations. β_0 is the expected value of Y_i when all X_{ki} are equal to zero.

Example for $k = 2$

Imagine we compare two schools, i and j

$$\begin{cases} Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} \\ Y_j = \beta_0 + \beta_1 X_{1j} + \beta_2 X_{2j} \end{cases}$$

$$\Rightarrow \begin{cases} Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} \\ Y_i + \Delta Y_i = \beta_0 + \beta_1 (X_{1i} + \Delta X_{1i}) + \beta_2 (X_{2j} + \Delta X_{2i}) \end{cases}$$

$$\Rightarrow \Delta Y_i = \beta_1 \Delta X_{1i} + \beta_2 \Delta X_{2i}$$

When X_2 is constant, i.e., $X_{2i} = X_{2j}$, then $\Delta X_{2i} = 0$ and therefore,

$$\Delta Y_i = \beta_1 \Delta X_{1i}$$

In general, holding all other regressors constant, $\frac{\Delta Y_i}{\Delta X_{1i}} = \beta_1$.

The OLS Estimator in Multiple Regression

As with simple regression, OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, ..., \hat{\beta}_k$ solve

$$\min_{b_0, b_1, b_2, \ldots, b_k} \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i} - ... - b_k X_{ki})^2$$

The derivative of the sum of squared prediction mistakes with respect to the jth regression coefficient, b_j, is

$$\frac{\partial}{\partial b_j} = \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i} - ... - b_k X_{ki})^2$$

$$\Leftrightarrow -2 \sum_{i=1}^{n} X_{ji} (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i} - ... - b_k X_{ki})$$

Combined, these yield the system of $k+1$ equations that, when set to 0, constitute the first-order conditions for the OLS estimator, $\hat{\boldsymbol{\beta}}$ — the $(k+1) \times 1$ dimensional vector of the $k+1$ estimates of the unknown regression coefficients

The OLS Estimator in Multiple Regression

The system of equation can be written as (with matricial notation)

$$\begin{aligned} -2_{k+1}\boldsymbol{X}'(\boldsymbol{Y}-\boldsymbol{X}\hat{\boldsymbol{\beta}}) &= \mathbf{0}_{k+1} \\ \Rightarrow \boldsymbol{X}'\boldsymbol{Y}-\boldsymbol{X}'\boldsymbol{X}\hat{\boldsymbol{\beta}} &= \mathbf{0}_{k+1} \\ \Rightarrow \hat{\boldsymbol{\beta}} &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y} \end{aligned}$$

Where

$$\boldsymbol{Y}_{n\times 1} = \begin{bmatrix} Y_1 \\ Y_2 \\ . \\ . \\ . \\ Y_n \end{bmatrix} \quad ; \quad \boldsymbol{X}_{n\times(k+1)} = \begin{bmatrix} 1 & X_{11} & X_{21} & \dots & X_{k1} \\ 1 & X_{12} & X_{22} & \dots & X_{k2} \\ . & . & . & \dots & . \\ . & . & . & \dots & . \\ . & . & . & \dots & . \\ 1 & X_{1n} & X_{2n} & \dots & X_{kn} \end{bmatrix} \text{ ; and } \quad \hat{\boldsymbol{\beta}}_{(k+1)\times 1} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ . \\ . \\ . \\ \hat{\beta}_k \end{bmatrix}$$

Example: The impact of Subsidized Meals on Test Scores

```
. reg testscore frpm_frac_s , robust

Linear regression                               Number of obs     =        500
                                                F(1, 498)         =     437.13
                                                Prob > F          =     0.0000
                                                R-squared         =     0.5019
                                                Root MSE          =     42.601

------------------------------------------------------------------------------
             |               Robust
   testscore |   Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  frpm_frac_s |  -154.8953     7.40853   -20.91   0.000    -169.4512   -140.3395
        _cons |   847.072     4.807364   176.20   0.000     837.6268    856.5172
------------------------------------------------------------------------------
```

Example: The impact of Subsidized Meals on Test Scores

```
. reg  testscore frpm_frac_s ed_frac_bachplus, robust

Linear regression                               Number of obs     =        500
                                                F(2, 497)         =     241.13
                                                Prob > F          =     0.0000
                                                R-squared         =     0.5221
                                                Root MSE          =     41.772
```

testscore	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
frpm_frac_s	-122.532	9.458738	-12.95	0.000	-141.116	-103.9479
ed_frac_bachplus	89.04404	19.42138	4.58	0.000	50.88591	127.2022
_cons	806.3163	9.468789	85.16	0.000	787.7125	824.9201

Goodness of Fit in Multiple Regression

Remember that:

$$\text{Actual} = \text{Predicted} + \text{Residual} \ (Y_i = \hat{Y}_i + \hat{u}_i)$$

In the previous lectures, we have seen that:

- RMSE = std. deviation of $\hat{u}_i$ *without* d.f. correction
- SER = std. deviation of $\hat{u}_i$ *with* d.f. correction
- R^2 = share of the variance of Y explained by X *without* d.f. correction

Now, we will introduced the adjusted R^2, written $\overline{R}^2$

$\Rightarrow$ $\overline{R}^2$ = share of the variance of Y explained by X *with* d.f. correction

Goodness of Fit in Multiple Regression

With multiple regressors,

- RMSE $= \sqrt{\frac{1}{n}\sum_{i=1}^{n}\hat{u}_i^2}$
- SER $= \sqrt{\frac{1}{n-(k+1)}\sum_{i=1}^{n}\hat{u}_i^2}$
- $R^2 = \frac{\sum_{i=1}^{n}(\hat{Y}_i-\overline{Y})^2}{\sum_{i=1}^{n}(Y_i-\overline{Y})^2} = 1 - \frac{\sum_{i=1}^{n}\hat{u}_i^2}{\sum_{i=1}^{n}(Y_i-\overline{Y})^2}$

BUT whenever a variable is added to the RHS, the sum of squared residuals is (almost systematically) smaller (the OLS finds the values of the coefficients that minimize the sum of squared residuals) so the $SSR \downarrow \Rightarrow R^2 \uparrow$

Goodness of Fit in Multiple Regression

The $\overline{R}^2$, "adjusted" R^2 corrects this by penalizing the inclusion of an additional regressor:

$$\overline{R}^2 = 1 - \frac{n-1}{n-k-1} \frac{\sum_{i=1}^{n} \hat{u}_i^2}{\sum_{i=1}^{n} (\hat{Y}_i - \overline{Y})^2}$$

Note that:

- $\overline{R}^2 < R^2$ because $\frac{n-1}{n-k-1} > 1$ but when $n \to \infty$ they become very close.
- Adding one regressor $\downarrow SSR$ but also $\uparrow \frac{n-1}{n-k-1}$
- $\overline{R}^2$ may be negative if the (many) regressors $\downarrow SSR$ but $\uparrow\uparrow\uparrow \frac{n-1}{n-k-1}$

However, including a variable in a multiple regression should be based on whether including that variable allows you to better identify the causal effect of interest.

The OLS Assumptions for Causal Inference in the Multiple Regression Model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + ... + \beta_k X_{ki} + u_i \qquad \text{with} \quad i = 1, \ldots, n$$

where $\beta_1, \ldots, \beta_k$ are causal effects and

1. u_i has a conditional mean of 0 given $X_{1i}, X_{2i}, \ldots, X_{ki}$; that is,

$$\mathbb{E}(u_i \mid X_{1i}, X_{2i}, \ldots, X_{ki}) = 0$$

2. $(X_{1i}, X_{2i}, \ldots, X_{ki}, Y_i)$ are i.i.d. draws from their joint distribution.
3. Large outliers are unlikely: $(X_{1i}, X_{2i}, \ldots, X_{ki}, Y_i)$ have nonzero finite fourth moments
4. There is no perfect multicollinearity.

$\Rightarrow$ If assumptions 1-4 hold, then in large samples the OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k$ are jointly normally distributed, and each $\hat{\beta}_j \sim \mathcal{N}(\beta_j; \sigma^2_{\hat{\beta}_j})$.

What is Multicollinearity?

Multicollinearity refers to the high correlation between two or more independent variables in a regression model.

- **Perfect Multicollinearity:** A linear relationship exists between independent variables (e.g., $X_3 = 0.5 \times$ Share w/ College Education).
- **High Multicollinearity:** High correlation but not a perfect linear relationship (e.g., ρ(Share w/ College Education; Median Income) $= 0.8755$).

Example: Perfect Multicolinearity

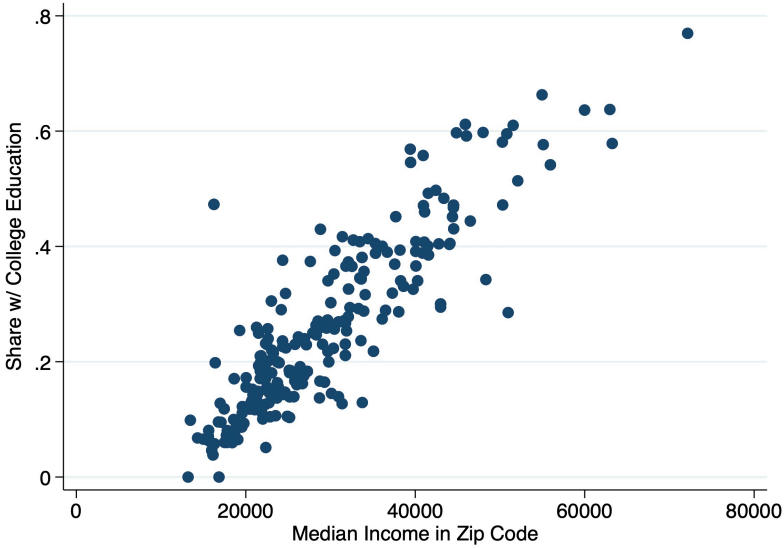
```
. ge X3 = ed_frac_bachplus/2

.  reg  testscore frpm_frac_s ed_frac_bachplus X3, robust
note: X3 omitted because of collinearity.

Linear regression                               Number of obs     =        500
                                                F(2, 497)         =     241.13
                                                Prob > F          =     0.0000
                                                R-squared         =     0.5221
                                                Root MSE          =     41.772
```

testscore	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
frpm_frac_s	-122.532	9.458738	-12.95	0.000	-141.116	-103.9479
ed_frac_bachplus	89.04404	19.42138	4.58	0.000	50.88591	127.2022
X3	0	(omitted)				
_cons	806.3163	9.468789	85.16	0.000	787.7125	824.9201

Example: High Multicolinearity



Example: High Multicolinearity

```
. reg testscore frpm_frac_s ed_frac_bachplus med_income_z , robust

Linear regression                               Number of obs     =        500
                                                F(3, 496)         =     162.85
                                                Prob > F          =     0.0000
                                                R-squared         =     0.5222
                                                Root MSE          =     41.811
```

testscore	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
frpm_frac_s	-123.2939	10.02643	-12.30	0.000	-142.9934	-103.5944
ed_frac_bachplus	94.213	26.47106	3.56	0.000	42.20376	146.2222
med_income_z	-.0001054	.000426	-0.25	0.805	-.0009423	.0007315
_cons	808.497	13.40335	60.32	0.000	782.1627	834.8314

The dummy variable trap

Suppose you have a set of multiple mutually exclusive and exhaustive variables — that is, there are multiple categories and every observation falls in one and only one category (Freshmen, Sophomores, Juniors, Seniors, Other).

If you include all these variables and a constant (in general, the intercept), you will have perfect multicollinearity — this is sometimes called the dummy variable trap (as it often occurs with dummy variables).

We can then decide to omit one of the groups (or the constant). In this case, the coefficients on the included binary variables represent the incremental effect of being in that category, relative to the base case of the omitted category.

Example: Omitted category 1/2

```
. ge ed_frac_bachless = 1 - ed_frac_bachplus

. reg  testscore frpm_frac_s ed_frac_bachplus ed_frac_bachless , robust
note: ed_frac_bachless omitted because of collinearity.

Linear regression                               Number of obs     =        500
                                                F(2, 497)         =     241.13
                                                Prob > F          =     0.0000
                                                R-squared         =     0.5221
                                                Root MSE          =     41.772
```

testscore	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
frpm_frac_s	-122.532	9.458738	-12.95	0.000	-141.116	-103.9479
ed_frac_bachplus	89.04404	19.42138	4.58	0.000	50.88591	127.2022
ed_frac_bachless	0	(omitted)				
_cons	806.3163	9.468789	85.16	0.000	787.7125	824.9201

Example: Omitted category 2/2

```
. reg  testscore frpm_frac_s ed_frac_bachplus ed_frac_bachless , robust nocons

Linear regression                               Number of obs     =        500
                                                F(3, 497)         =   55420.21
                                                Prob > F          =     0.0000
                                                R-squared         =     0.9970
                                                Root MSE          =     41.772

------------------------------------------------------------------------------
             |               Robust
  testscore  | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
   frpm_frac_s |  -122.532   9.458738   -12.95   0.000    -141.116   -103.9479
ed_frac_bachplus |  895.3603   11.95777    74.88   0.000    871.8663    918.8543
ed_frac_bachless |  806.3163   9.468789    85.16   0.000    787.7125    824.9201
------------------------------------------------------------------------------
```

Control Variables in Multivariate Analysis

Definition

A control variable W is a regressor included to hold constant factors that, if neglected, could lead the estimated causal effect of interest to suffer from omitted variable bias.

This definition means that

1. We need to modify the OLS assumptions such that the OLS estimator of the effect of interest is unbiased, but the OLS coefficients on control variables are, in general, biased and do not have a causal interpretation.
2. A good control variable is one which, when included in the regression, makes the error term uncorrelated with the variable of interest.
3. Holding constant the control variable(s), the variable of interest is "as if" randomly assigned.
4. Among individuals (entities) with the same value of the control variable(s), the variable of interest is uncorrelated with the omitted determinants of Y

Conditional Mean Independance

- Because a control variable is correlated with an omitted causal factor, the first OLS Assumption, $\mathbb{E}(u_i \mid X_{1i}, X_{2i}, \ldots, X_{ki}) = 0$ does not hold anymore
- We need a mathematical condition for what makes an effective control variable. This condition is the *conditional mean independence*
- Given the control variable, the mean of u_i does not depend on the variable of interest
- Question: Conditional on sharing the same exposure to adults' educational attainment, is the share of subsidized meals as good as random?

The OLS Assumptions for Causal Inference in the Multiple Regression Model with Control Variables

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + ... + \beta_k X_{ki} + \beta_{k+1} W_{1i} + \beta_{k+2} W_{2i} + ... + \beta_{k+r} W_{ri} + u_i \qquad \text{with} \quad i = 1, \ldots, n$$

where $\beta_1, \ldots, \beta_k$ are causal effects, W_{ji} are control variables, and

1. u_i has a conditional mean that does not depend on the X's given the W's; that is,
$$\mathbb{E}(u_i \mid X_{1i}, X_{2i}, \ldots, X_{ki}, W_{1i}, W_{2i}, \ldots, W_{ri}) = \mathbb{E}(u_i \mid W_{1i}, W_{2i}, \ldots, W_{ri})$$
2. $(X_{1i}, X_{2i}, \ldots, X_{ki}, W_{1i}, W_{2i}, \ldots, W_{ri}, Y_i)$ are i.i.d. draws from their joint distribution.
3. Large outliers are unlikely: $(X_{1i}, X_{2i}, \ldots, X_{ki}, W_{1i}, W_{2i}, \ldots, W_{ri}, Y_i)$ have nonzero finite fourth moments
4. There is no perfect multicollinearity.

$\Rightarrow$ If assumptions 1-4 hold, then in large samples the OLS estimators of interest, $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k$ are jointly normally distributed, and each $\hat{\beta}_j \sim \mathcal{N}(\beta_j; \sigma^2_{\hat{\beta}_j})$ BUT $\hat{\beta}_{k+1}, \hat{\beta}_{k+2}, \ldots, \hat{\beta}_{k+r}$ might be biased!

Material I

- **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapter 6.